

Sound Direction Detection

PROBLEM STATEMENT

Sound Direction Detection

Problem Statement:

Develop a hybrid analog and digital system for real-time directional detection of sound using basic electronic components. The system will utilize multiple microphones arranged in a specific geometry to capture incoming sound waves and analyze the time difference of arrival (TDOA) to estimate the direction of the sound source. The project aims to create a cost-effective, low-power, and hardware-efficient solution for applications such as environmental monitoring, security, and assistive technology.

APPROACH / DESIGN/ METHODOLOGY

Microphone Array Setup

- Arrange multiple microphones in a square pattern to capture sound from different directions.
- Ensure proper spacing to maximize time difference of arrival (TDOA) for accurate direction detection.

Analog Signal Processing

- Amplify the microphone signals using operational amplifiers.
- Use a peak detector to smooth out sudden impulses .
- Convert the processed signal into a negative trigger using a BJT circuit.

Pulse Generation

- Use a 555 timer circuit to generate a uniform 50 ms pulse when a sound impulse is detected.
- Implement four such circuits, one for each microphone channel.

Time Difference Measurement

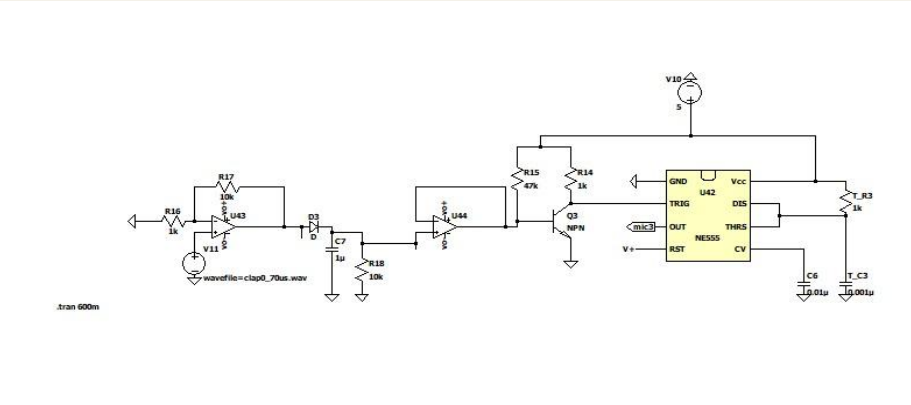
- Use six flip-flops to determine which pulse arrives first and which arrives last.

Digital Processing for Direction Detection

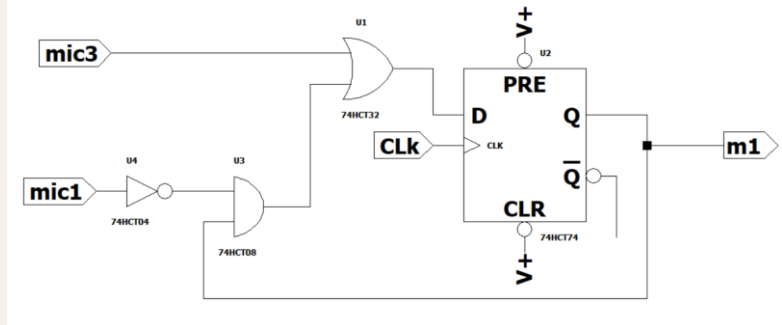
- Design a digital circuit to interpret the flip-flop outputs and derive the sound direction.

CIRCUIT DIAGRAM /ALGORITHM

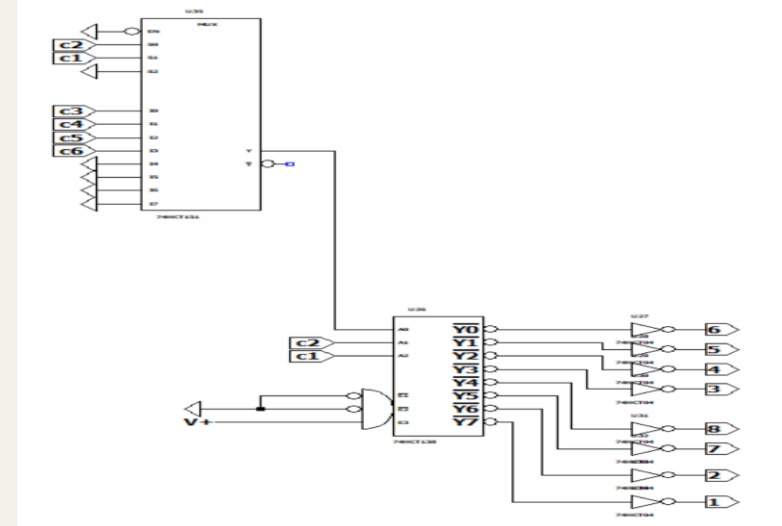
Input analog circuit – sound to pulse converter x 4



SR Flip –Flop x 6 ($4C_2$)



Output Decoder



OUTPUT LOGIC

SR 1	SR 1	Output Led
M1<M3	M2<M4	
0	0	5,6
0	1	3,4
1	0	7,8
1	1	1,2

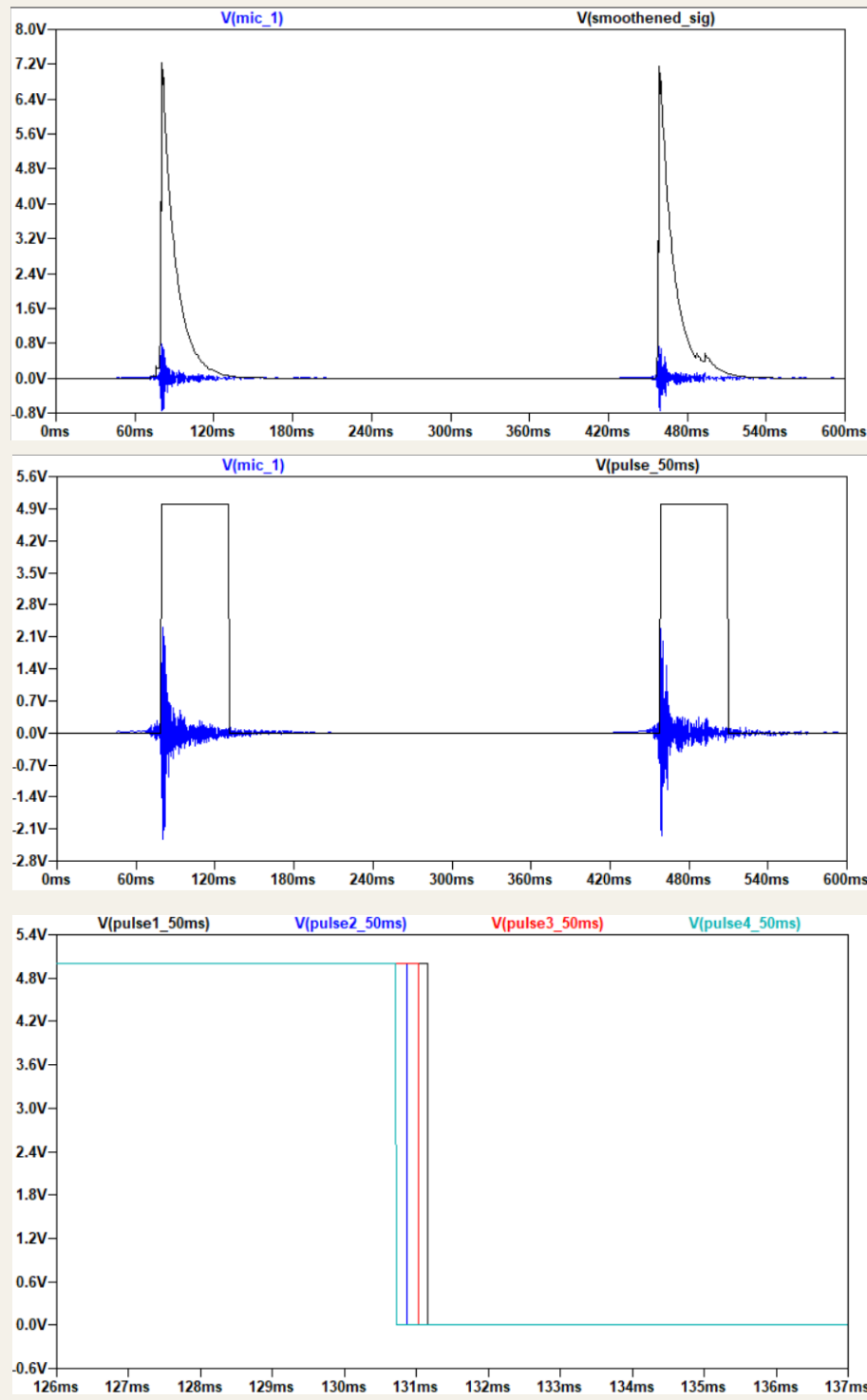
SR 3	
M3<M4	
0	6
1	5

SR 4	
M2<M3	
0	4
1	3

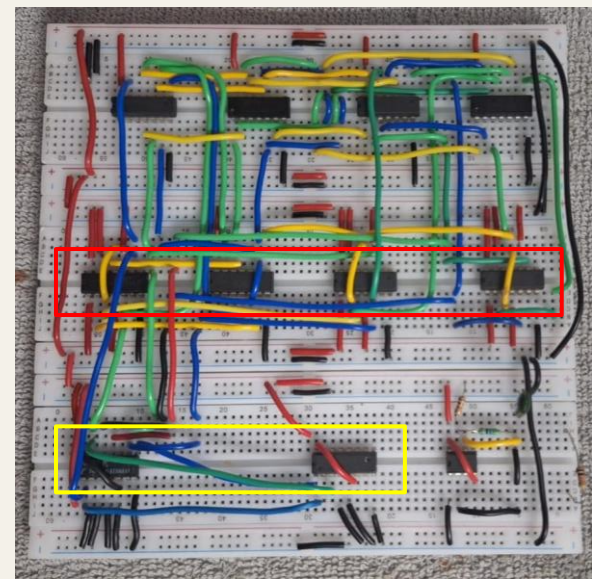
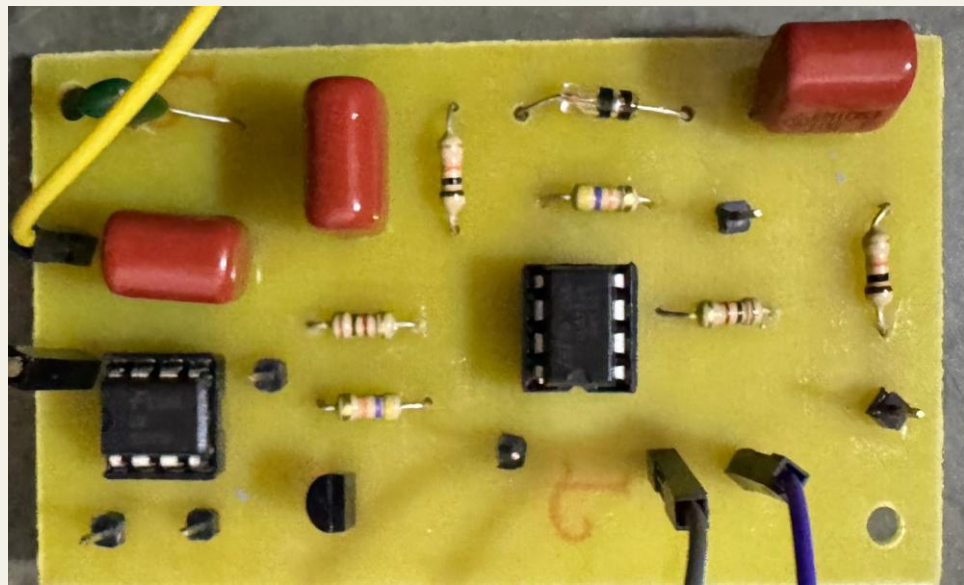
SR 5	
M4<M1	
0	8
1	7

SR 6	
M1<M2	
0	2
1	1

SIMULATION RESULTS

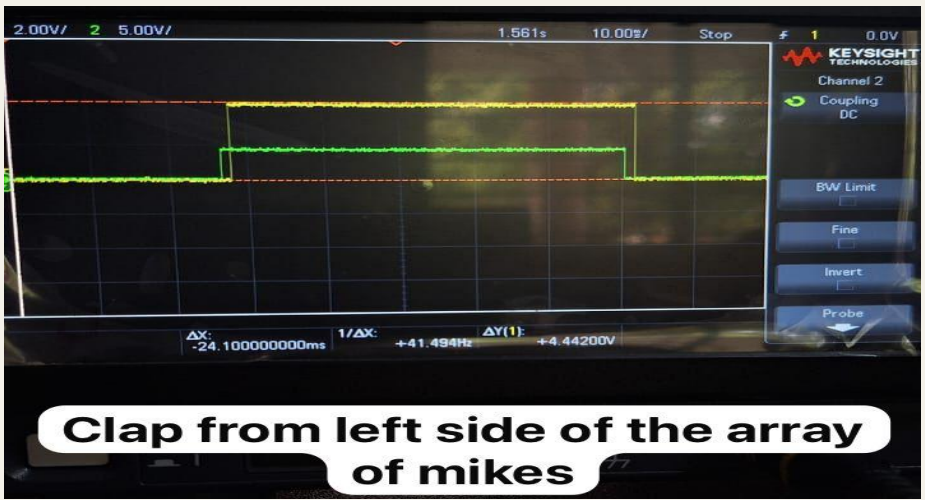


WORKING MODEL PICTURES

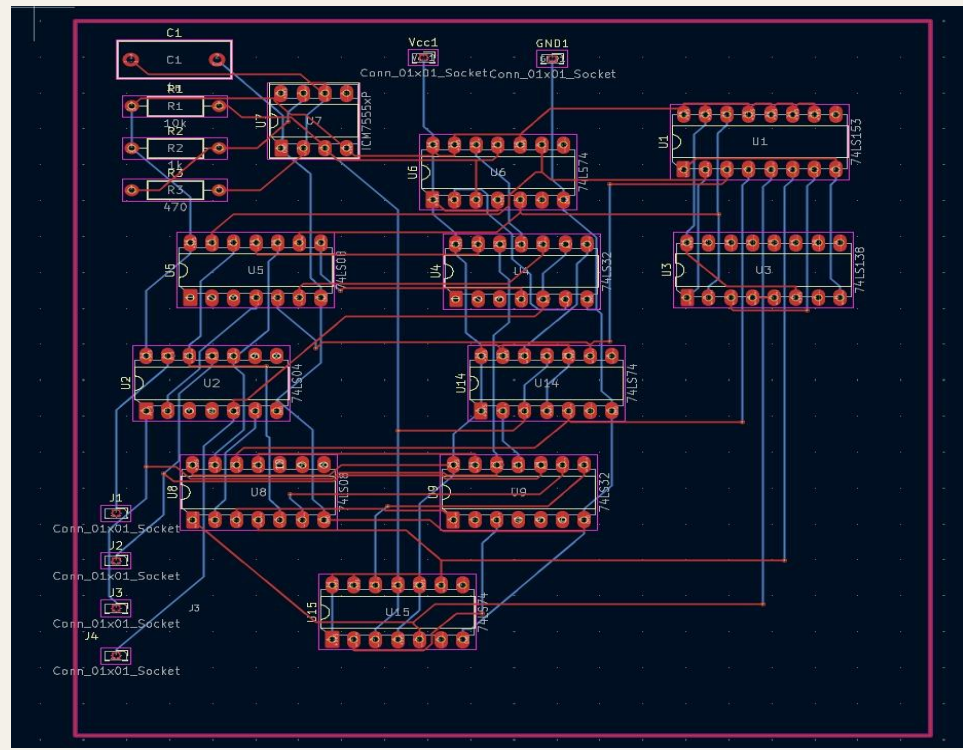
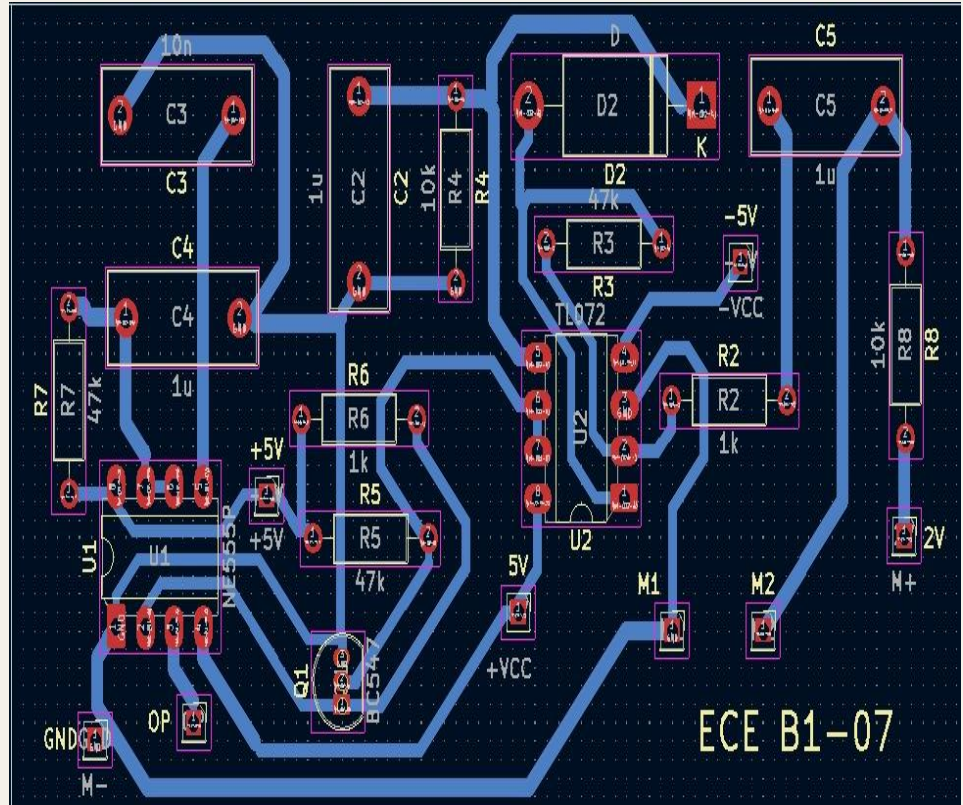


SR -Flipflop
Decoder circuit

RESULTS/ CALIBRATION/ ERROR GRAPH



PCB LAYOUT



REFERENCES

1. A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed., Oxford University Press, 2015, Ch. 2-3.
2. A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed., Oxford University Press, 2015, Ch. 4-5.
3. P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed., Cambridge University Press, 2015, Ch. 1.
4. <https://www.soberton.com/wp-content/uploads/2019/05/EM-6050-14-Feb-2019.pdf>